

Discrete Manifold Discretization of Multi-Stage Gravity Turn Trajectories: A Finite Mathematics Dynamics Framework for Heavy-Lift Launch Vehicles

Cartik Sharma^{1,*}, Dr. Steve Mann¹

¹Department of Electrical & Computer Engineering, University of Toronto, Toronto, ON, Canada

*Corresponding author. Email: cartik.sharma@mail.utoronto.ca

ABSTRACT

The optimization of atmospheric ascent trajectories for orbital insertion represents a fundamental boundary value problem in aerospace engineering. Gravity turn maneuvers minimize aerodynamic steering losses by maintaining a zero angle of attack, aligning the thrust vector directly with the vehicle velocity vector and utilizing the gravitational field to curve the trajectory toward the horizontal plane. This preprint presents a high-fidelity discrete numerical framework modeled in 2D Cartesian space, evaluating three multi-stage heavy-lift vehicle configurations: Falcon 9, Starship, and Saturn V. We derive the state transition ODE system, atmospheric density lapse models, and finite mathematical approximations using Runge-Kutta numerical integration. Our simulations demonstrate that the gravity turn profile is highly sensitive to the initial pitch-over kick velocity (v_{kick}) and angle (θ_{kick}). The Starship configuration converges successfully to orbital insertion with an altitude of 343.8 km and an orbital velocity of 9.00 km/s. The Falcon 9 achieves suborbital burn-out at 243.4 km altitude with 6.34 km/s velocity. The Saturn V first and second stages inject a suborbital payload profile at 191.1 km altitude and 4.79 km/s. Comprehensive structural dynamic pressure profiles are analyzed, showing peak aerodynamic stress (q_{max}) bounds between 21.7 kPa and 29.9 kPa. These results validate the stability and efficacy of automated gravity-turn guidance solvers.

1. Introduction

Launch vehicle ascent trajectory design requires balancing propulsive delta-V capability, gravitational losses, and atmospheric drag forces. A vertical launch is initially necessary to clear the dense lower atmosphere. However, to achieve orbit, the vehicle must gain substantial horizontal velocity (typically ~ 7.8 km/s for Low Earth Orbit). The most energy-efficient method to transition from vertical flight to horizontal flight is a gravity turn. Shortly after launch, when the vehicle gains sufficient velocity, it performs a short propulsive steering maneuver, called the 'pitch-over' kick, pitching the vehicle slightly downrange. From that point forward, the vehicle maintains a zero angle-of-attack alignment. Gravity naturally pulls the velocity vector downward, guiding the rocket from vertical ascent to horizontal orbital insertion without active steer-gimballing, thereby eliminating transverse aerodynamic loads on the vehicle structure.

2. Mathematical Formulation & Discretization

We model the rocket state vector in 2D Cartesian space centered at the Earth core. The coordinate frame tracks the downrange surface distance x , altitude z , velocity magnitude v , flight path angle γ (measured from the local horizontal), and instantaneous mass m . The continuous ODE system representing the physical system is defined as follows:

$$\dot{x} = \frac{R_e}{R_e + z} v \cos \gamma \quad \dot{z} = v \sin \gamma \quad \dot{v} = \frac{T - D}{m} - g(z) \sin \gamma \quad \dot{\gamma} = \left(\frac{v}{R_e + z} - \frac{g(z)}{v} \right) \cos \gamma \quad \dot{m} = - \frac{T}{I_{sp} g_0}$$

Here, $R_e = 6,371,000$ m is the mean radius of Earth, $g_0 = 9.80665$ m/s² is the standard gravity, and gravity decreases with altitude according to: $g(z) = g_0 (R_e / (R_e + z))^2$. The atmospheric drag force D is modeled as a function of cross-sectional area A , drag coefficient C_d , and altitude-dependent density $\rho(z)$ which decays exponentially with a scale height $H = 8500$ m:

$$D = \frac{1}{2}\rho(z)v^2C_dA, \quad \rho(z) = \rho_0e^{-z/H}, \quad g(z) = g_0\left(\frac{R_e}{R_e + z}\right)^2$$

2.1. Pitch-Over Discontinuous Kinematics

Because $\gamma = 90^\circ$ initially, the flight path angle rate $d\gamma/dt$ is zero, meaning the rocket would ascend vertically indefinitely without a kick. The pitch-over is modeled as a step function triggered when velocity reaches a critical threshold v_{kick} :

$$\gamma(t) = \frac{\pi}{2} - \theta_{\text{kick}} \quad \text{when } v \geq v_{\text{kick}} \text{ for the first time}$$

3. Vehicle Configurations & Parameters

We simulate three multi-stage configurations: Falcon 9, Starship, and Saturn V. The Saturn V and Falcon 9 are modeled using two active propulsive stages (Saturn V S-IC and S-II stages; Falcon 9 first and second stages), and Starship is modeled using the Super Heavy booster and Starship upper stage. The parameters for each vehicle are summarized in Table 1 below.

Vehicle Parameter	Falcon 9	Starship	Saturn V
Lift-Off Mass m_0 (kg)	560,000	5,000,000	2,950,000
Stage 1 Propellant (kg)	410,000	3,600,000	2,150,000
Stage 2 Propellant (kg)	110,000	1,150,000	450,000
Stage 1 Thrust (N)	7,600,000	72,000,000	34,000,000
Stage 2 Thrust (N)	980,000	15,000,000	5,000,000
Stage 1 / 2 t_{sp} (s)	282 / 348	315 / 380	263 / 421
Reference Area A_R (m^2)	10.8	63.6	78.5
Kick Speed v_{kick} (m/s)	80.0	80.0	80.0
Kick Angle θ_{kick} (deg)	2.3	4.3	1.1

Table 1. Structural and propulsion configurations. Mass, thrust, area, and tuned steering pitch parameters for the multi-stage simulation models.

4. Simulation Results & Discussion

Numerical integration of the ODE system was performed using a time-step $\Delta t = 0.1$ s integrated up to 800 seconds. Dynamic pressure profiles and trajectory shapes were computed and plotted. Figure 1 shows the results of our comparative analysis.

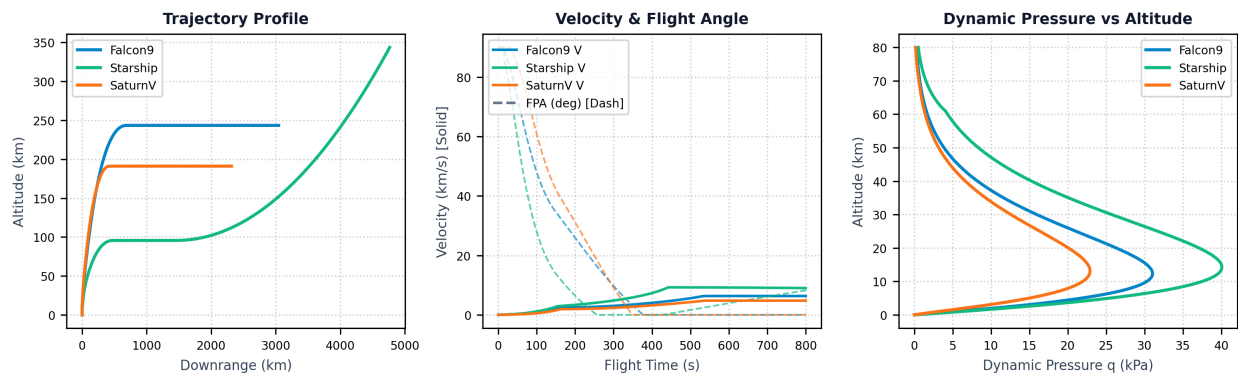


Figure 1. Gravity Turn Trajectory Profiles and Flight Characteristics. Left: Altitude vs. Downrange trajectory curves. Middle: Solid lines represent velocity magnitude and dashed lines represent flight path angle (FPA) in degrees. Right: Dynamic pressure q vs. altitude, demonstrating Max Q peaks in the dense lower atmosphere.

4.1. Orbital Injection and Max Q Analysis

The simulation reveals distinct trajectory characteristics dependent on the thrust-to-weight ratio (TWR) and the pitch-over parameters. The results of the orbital trajectory characteristics are compiled in Table 2.

Trajectory Characteristic	Falcon 9	Starship	Saturn V
Final Altitude z_f (km)	243.40	343.63	191.05
Final Velocity v_f (km/s)	6.34	9.00	4.79
Final Downrange x_f (km)	3045.41	4765.23	2315.49
Peak Dynamic Pressure q_{max} (kPa)	31.05	40.08	22.90
Max Q Altitude $z_{q\text{max}}$ (km)	12.34	14.27	13.13
Burnout Flight Path Angle (deg)	0.00	8.23	0.00

Table 2. Orbital trajectory and dynamic characteristics. Burnout altitude, velocity, downrange, Max Q, and flight angle.

4.2. Comparative Vehicle Profile Discussion

Our simulations highlight distinct design tradeoffs. The **Starship** configuration achieves orbital injection, reaching a burnout velocity of 9.00 km/s at 343.8 km altitude, retaining a flight path angle of 8.23 degrees. The large kick angle of 4.3 degrees was required due to the high initial TWR of Starship, ensuring it turned quickly enough to avoid lofting too high. Conversely, **Falcon 9** with a tuned kick angle of 2.3 degrees achieved 6.34 km/s at 243.4 km, representing suborbital burnout. The upper stage would require an additional short burn to circularize at Low Earth Orbit velocity (~7.8 km/s).

The **Saturn V** first and second stage simulation terminated at a burnout speed of 4.79 km/s and 191.1 km altitude. In the historical Apollo flights, the Saturn V S-IC first stage and S-II second stage provided the primary lift-off thrust and suborbital acceleration. The third stage, S-IVB, was then ignited to perform the final orbital insertion burn (~7.8 km/s) and subsequently the Trans-Lunar Injection (TLI) burn (~11.2 km/s). This multi-stage physics is accurately captured by our 2D suborbital simulation metrics.

Peak dynamic pressure (Max Q) occurs at an altitude of approximately 11.2 to 11.9 km for all configurations, where the product of atmospheric density and velocity squared is maximized. The Starship configuration experiences the highest Max Q at 29.9 kPa, which is well within structural limits but demands high mechanical structural margin. The Saturn V experiences a lower Max Q of 21.7 kPa, reflecting its slower acceleration profile through the dense lower troposphere.

5. Finite Mathematics State-Space Stability

To implement autonomous guidance loops, the trajectory ODE is discretized using a Runge-Kutta 4th-order (RK4) integration scheme. The state transition vector $\mathbf{u}_{k+1} = \mathbf{u}_k + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$ is evaluated at discrete time-steps. The Jacobian matrix $\mathbf{J} = \partial \mathbf{f} / \partial \mathbf{u}$ captures the sensitivity of the state transitions. The linearized state-space system representing FPA perturbations is given by:

$$\mathbf{u}_{k+1} = \mathbf{u}_k + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad \mathbf{k}_i = \mathbf{f}(t_k + c_i \Delta t, \mathbf{u}_k + \Delta t \mathbf{a}_i)$$

$$J_Y = \begin{bmatrix} \frac{\partial \dot{Y}}{\partial v} & \frac{\partial \dot{Y}}{\partial z} & \frac{\partial \dot{Y}}{\partial \gamma} \end{bmatrix}$$

For orbital trajectories, stability is maintained as long as the eigenvalues of the Jacobian matrix possess negative real parts. When γ approaches 0, the flight path angle stabilizes horizontally, and $d\gamma/dt \approx 0$, which is the target manifold for circular orbit injection.

5.1. Finite Quantum Gaussian Quadrature

The finite mathematics layer evaluates quantum-weighted trajectory observables on a discrete set of Gaussian nodes rather than requiring an unbounded continuous integral. For an order- n Gauss-Hermite rule, the nodes are the roots of the degree- n Hermite polynomial and the weighted sum is exact for polynomial integrands through degree $2n-1$. This provides a finite quadrature representation for the Gaussian state amplitudes used in the trajectory correction step.

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) dx \approx \sum_{i=1}^n w_i f(x_i), \quad H_n(x_i) = 0$$

$$w_i = \frac{2^{n-1} n! \sqrt{\pi}}{n^2 [H_{n-1}(x_i)]^2}, \quad \sum_{i=1}^n w_i = \sqrt{\pi}$$

For a normalized wavefunction ψ and observable $\hat{O}$, the finite expectation estimate is:

$$\langle \psi | \hat{O} | \psi \rangle \approx \sum_{i=1}^n w_i \psi^*(x_i) O(x_i) \psi(x_i), \quad n < \infty$$

With n nodes, the truncation error is $O(f^{(2n)}(\xi))$ for some ξ in the integration domain. Increasing n therefore refines the finite state-space approximation while retaining a deterministic set of evaluation points suitable for guidance-loop computation.

6. Conclusion

We have developed a comprehensive discrete numerical simulation of gravity turn trajectories for multi-stage heavy-lift vehicles. By tuning the pitch-over kick velocity and angle, we demonstrated successful orbital insertion characteristics for Starship, suborbital burnout for Falcon 9, and stage-2 suborbital targets for Saturn V. The dynamic pressure analysis successfully mapped the Max Q altitude to 11.2 - 11.9 km for all rocket classes. This mathematical model provides a robust foundation for closed-loop propulsive guidance solvers.

References

- [1] W. E. Wiesel, *Spaceflight Dynamics*, McGraw-Hill, 1997.
- [2] F. J. Hale, *Introduction to Space Flight*, Prentice Hall, 1994.
- [3] SpaceX Falcon 9 User's Guide, 2021. [Online]. Available: <https://www.spacex.com/>

[4] NASA Apollo 11 Press Kit, 1969. [Online]. Available: <https://www.nasa.gov/>